A/B TESTING CASE STUDY

White vs Black Website Background

End-to-End Project Documentation

Completed by

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Tools: Python • Pandas • SciPy • Statsmodels • Seaborn • Matplotlib

Dataset: ab_testing.csv

1. Executive Summary

This project investigates whether website background color is associated with differences in user engagement and conversion. Group A is mapped to White (control) and Group B to Black (treatment). The supplied dataset is a synthetic practice dataset representing visits to an imaginary retail website. Time spent is compared using Welch's independent two-sample t-test and conversion rates are compared using a two-proportion z-test.

Outcome	White (A)	Black (B)	Interpretation
Visitors	2,519	2,481	Similar sample sizes
Conversions	136	349	More recorded conversions in Black group
Conversion rate	5.40%	14.07%	+8.67 percentage points for Black
Relative conversion lift	Baseline	160.55%	Relative to White conversion rate
Mean time spent	241.73	243.30	Difference +1.57 units
Median time spent	241	244	Descriptive comparison
Welch time test	—	—	$t = 0.4695$; $p = 0.6387$; not significant
Conversion z-test	—	—	Reported $z = 10.3545$; $p = 3.99 \times 10^{-25}$

Main conclusion: the Black group has a much higher observed conversion rate, but average time spent does not differ significantly. Because the data is synthetic and assignment/randomization is unverified, these findings are practice results not proof that background color caused the difference.

2. What Is This Project About?

A/B testing compares two versions of a product or experience against a defined outcome. Here, the intended change is the website background color. The business question is: Do users spend more time on a website with a white or black background and does either version have a higher conversion rate?

Project element	Definition
Group A / control	White background (default version)
Group B / treatment	Black background (new version being tested)
Primary metric	Time spent per user/session
Secondary metric	Conversion rate = conversions ÷ users in group
Significance level	$\alpha = 0.05$
Dataset status	Synthetic practice data; not verified real customer behavior

A conversion means a desired action recorded as Yes/No. The CSV does not specify whether that action is a purchase, signup, or another event.

3. Dataset Description

The supplied CSV contains 5,000 rows and 8 columns. Its accompanying description identifies it as a self-generated synthetic dataset made using NumPy random sampling, representing an imaginary retail website.

Column	Meaning	Use
User ID	User identifier	Reference only
Group	A or B	Map A → White, B → Black
Page Views	Pages viewed	Optional engagement measure
Time Spent	Session duration; dataset notes specify seconds	Primary outcome
Conversion	Yes / No	Secondary outcome

Device	Device type	Possible segmentation
Location	Location in UK	Possible segmentation

Important assumption: the CSV has a Group column but no Background column. The mapping A = White and B = Black comes from the project brief and should be verified against the dataset-generation process.

4. Objectives

- Compare sample sizes for both groups.
- Calculate average and median time spent.
- Test whether the average time-spent difference is statistically significant.
- Calculate conversion counts and conversion rates.
- Test whether conversion rates differ significantly.
- Translate the results into a cautious stakeholder recommendation.

5. End-to-End Workflow

1 . Define the question: Set the primary metric (time spent) and secondary metric (conversion) before analysis.

2 . Load data: Use pandas . read_csv() to import the CSV.

3 . Inspect: Check shape, columns, missing values, duplicates, data types and group labels.

4 . Clean: Standardize group labels, convert time to numeric and handle missing/duplicate records carefully.

5 . Map groups: Create Background with A = White and B = Black.

6 . Explore: Calculate group counts, means, medians and standard deviations.

7 . Test time: Use Welch's two-sample t-test, which does not assume equal group variances.

8 . Test conversions: Use a two-proportion z-test to compare conversion rates.

9 . Visualize: Create a mean comparison chart and a boxplot.

10 . Recommend: Interpret p-values, effect sizes, business relevance and limitations.

6. Hypotheses and Statistical Tests

6.1 Time Spent

H_0: μBlack = μWhite — population mean time spent is equal.

H_1: μBlack ≠ μWhite — population mean time spent differs.

Test: two-sided Welch independent two-sample t-test, using scipy.stats.ttest_ind(black, white, equal_var=False). Compare p with α = 0.05.

6.2 Conversion Rate

H_0: pBlack = pWhite — population conversion proportions are equal.

H_1: pBlack ≠ pWhite — population conversion proportions differ.

Test: two-proportion z-test. The supplied analysis reports z = 10.3545 and p = 3.99×10^{-25}. This is below 0.05 and indicates a statistically significant difference under the test assumptions.

A p-value is the probability, assuming the null hypothesis and test assumptions are true, of seeing a result at least as extreme as the observed one. It is not the probability that the null hypothesis is true.

7. Environment and Setup

Suggested folder structure:

```
Website_AB_Testing/
├── ab_testing.csv
├── ab_test_analysis.py
├── average_time_comparison.png
└── time_spent_distribution.png
```

Install the libraries in the VS Code terminal:

```
python -m pip install pandas scipy matplotlib seaborn statsmodels
```

Run the script from the project folder:

```
python ab_test_analysis.py
```

The CSV should be in the same working directory as the Python script, or the code should use the correct full/relative path.

8. Complete Python Code

This script performs data inspection, cleaning, descriptive analysis, a Welch t-test, charts, conversion summaries and a two-proportion z-test.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind
from statsmodels.stats.proportion import proportions_ztest

# 1. Load and inspect
df = pd.read_csv("ab_testing.csv")
print(df.head())
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
print("Missing values:\n", df.isnull().sum())
print("Exact duplicate rows:", df.duplicated().sum())

# 2. Clean
df = df.drop_duplicates()
df["Group"] = df["Group"].astype(str).str.strip().str.upper()
df["Time Spent"] = pd.to_numeric(df["Time Spent"], errors="coerce")
df = df.dropna(subset=["Group", "Time Spent"])
df = df[df["Group"].isin(["A", "B"])]

# Assumption from brief: A = White, B = Black
df["Background"] = df["Group"].map({"A": "White", "B": "Black"})

# 3. Descriptive statistics
print("\nGroup sizes:\n", df["Background"].value_counts())
summary = df.groupby("Background")["Time Spent"].agg(
    ["count", "mean", "median", "std"]
)
print("\nTime-spent summary:\n", summary.round(2))

white = df.loc[df["Background"] == "White", "Time Spent"]
black = df.loc[df["Background"] == "Black", "Time Spent"]
difference = black.mean() - white.mean()
print("Mean difference (Black - White):", round(difference, 2))
print("Relative mean difference (%):", round(difference / white.mean() * 100, 2))
```

```
# 4. Welch's two-sample t-test
t_stat, p_time = ttest_ind(black, white, equal_var=False)
print("\nWelch t-statistic:", round(t_stat, 4))
print("Time-spent p-value:", p_time)
print("Time conclusion:",
      "Significant difference" if p_time < 0.05
      else "No statistically significant difference")

# 5. Visualize time spent
sns.set_theme(style="whitegrid")
sns.barplot(data=df, x="Background", y="Time Spent",
            estimator="mean", errorbar=("ci", 95))
plt.title("Average Website Time: White vs Black")
plt.xlabel("Website Background")
plt.ylabel("Time Spent (seconds, per dataset description)")
plt.tight_layout()
plt.savefig("average_time_comparison.png", dpi=300)
plt.show()

sns.boxplot(data=df, x="Background", y="Time Spent")
plt.title("Distribution of Time Spent by Background")
plt.xlabel("Website Background")
plt.ylabel("Time Spent (seconds, per dataset description)")
plt.tight_layout()
plt.savefig("time_spent_distribution.png", dpi=300)
plt.show()

# 6. Conversion analysis (clean separately to retain valid Yes/No rows)
conv = pd.read_csv("ab_testing.csv")
conv["Group"] = conv["Group"].astype(str).str.strip().str.upper()
conv["Conversion"] = conv["Conversion"].astype(str).str.strip().str.lower()
conv = conv[
    conv["Group"].isin(["A", "B"]) &
    conv["Conversion"].isin(["yes", "no"])
]
conv["Background"] = conv["Group"].map({"A": "White", "B": "Black"})
conv["Converted"] = conv["Conversion"].eq("yes")

conv_summary = conv.groupby("Background")["Converted"].agg(
    conversions="sum", visitors="count", rate="mean"
)
conv_summary["rate_pct"] = conv_summary["rate"] * 100
print("\nConversion summary:\n", conv_summary.round(4))

# Compare Black vs White
b = conv_summary.loc["Black"]
w = conv_summary.loc["White"]
z_stat, p_conversion = proportions_ztest(
    [int(b["conversions"]), int(w["conversions"])],
    [int(b["visitors"]), int(w["visitors"])]
)
print("Z-statistic:", round(z_stat, 4))
print("Conversion p-value:", p_conversion)
print("Absolute difference (percentage points):",
      (b["rate"] - w["rate"]) * 100)
print("Relative lift (%):",
      (b["rate"] - w["rate"]) / w["rate"] * 100)
print("Conversion conclusion:",
      "Significant difference" if p_conversion < 0.05
      else "No statistically significant difference")
```

9. Results and Findings

9.1 Time Spent

Metric	White (A)	Black (B)
Visitors with valid time	2,519	2,481
Mean	241.73	243.30
Median	241.00	244.00
Mean difference (Black − White)	—	+1.57
Relative difference vs White	—	+0.65%
Welch t-statistic	—	0.4695
Welch p-value	—	0.6387

Interpretation: Black has a 1.57-unit higher observed mean, but $p = 0.6387 > 0.05$. Fail to reject the null hypothesis. The data does not provide sufficient evidence of a difference in population mean time spent; this does not prove the backgrounds are equivalent.

9.2 Conversion

Metric	White (A)	Black (B)
Visitors	2,519	2,481
Conversions	136	349
Conversion rate	5.40%	14.07%
Absolute difference	—	+8.67 percentage points
Relative lift	Baseline	+160.55%
Reported z-statistic	—	10.3545
Reported p-value	—	3.99×10^{-25}

Interpretation: the observed conversion rate is 14.07% for Black and 5.40% for White. The supplied analysis reports $z = 10.3545$ and $p = 3.99 \times 10^{-25}$, indicating a statistically significant difference under the test assumptions. Relative lift is approximately 160.56%; absolute improvement is 8.67 percentage points. These sample results do not guarantee future gains.

10. What Did We Find?

- Black: 349 conversions among 2,481 users; White: 136 among 2,519.
- Observed conversion rate: Black 14.07% versus White 5.40%.
- Black is 8.67 percentage points higher, about 160.55% relative lift compared with White.
- Mean time: Black 243.30 versus White 241.73; difference 1.57 units.
- Time-spent test: $t = 0.4695$, $p = 0.6387$; not statistically significant.
- Overall: a strong observed conversion-rate association, but no statistically significant difference in average time spent.

11. Stakeholder Decision and Recommendation

Recommendation: treat the black background as a promising candidate for a controlled follow-up experiment, not as an automatic permanent rollout.

- Verify group assignment and data provenance; the dataset is synthetic and cannot establish actual customer impact.
- Run a real randomized experiment with eligible users assigned randomly to White or Black.

- Predefine conversion as the primary business outcome if conversions are the business goal; keep time spent as a secondary metric.
- Keep page content, layout, pricing, calls to action, load performance and traffic sources consistent.
- Check accessibility, readability, contrast, focus indicators, mobile rendering and user preferences.
- Monitor guardrails such as bounce rate, page errors, page load time, customer satisfaction and refunds/cancellations if relevant.
- Use confidence intervals and the value of a conversion to estimate business impact; do not convert relative lift directly into revenue without supporting data.
- Roll out gradually only if a real test reproduces the result and guardrail metrics remain acceptable.

12. Limitations

- Synthetic data may not reflect real customer behavior.
- The CSV contains Group A/B, not a recorded Background column; the mapping is assumed from the project brief.
- Random assignment and comparable exposure are not verified, so causal claims are not justified.
- Device and location differences may affect outcomes; segment checks are a useful next step.
- Multiple outcome tests can increase false-positive risk; define the primary outcome in advance.
- Statistical significance is not the same as business value or accessibility.
- The dataset description states Time Spent is in seconds; verify source documentation before publishing the unit.
- No revenue or margin column is provided, so revenue impact cannot be calculated directly.

13. Quality Checks Before Publishing

- Confirm A = White and B = Black with the dataset source.
- Check missing values, unexpected labels, exact duplicates and repeated User IDs.
- Confirm that duplicate removal is appropriate for the data-generation design.
- Inspect the time-spent distribution and outliers.
- Re-run both tests on the same validated data and record package versions.
- Report sample sizes, absolute effect, relative lift, p-values and limitations.

15. Glossary

Term	Meaning
A/B test	Compare two versions against a defined outcome.
Control group	Baseline version; White (A).
Treatment group	Changed version; Black (B).
Conversion rate	Conversions divided by users in a group.
Mean / median	Average / middle value after sorting.
Null hypothesis	Default claim that there is no difference.
p-value	How unusual the observed result would be if the null and test assumptions held.
Statistical significance	Evidence against the null at a chosen threshold; not proof of causation.
Percentage points	Direct subtraction of two percentages.
Relative lift	(New rate − baseline rate) ÷ baseline rate.

16. Final Conclusion

This project demonstrates an end-to-end analytics workflow: define the business question, inspect and clean data, compare groups, run statistical tests, visualize outcomes and recommend next steps. In the synthetic dataset, Black has a substantially higher observed conversion rate, while the difference in average time spent is not statistically significant. The responsible stakeholder decision is to validate the conversion finding in a randomized real-world experiment and check accessibility and user experience before rollout.

Appendix: Quick Answers

Question	Answer
Does Black have a higher observed conversion rate?	Yes: 14.07% vs 5.40%.
Is reported conversion difference significant?	Yes; reported $z = 10.3545$, $p = 3.99 \times 10^{-25}$.
Is average time higher for Black?	Descriptively yes: 243.30 vs 241.73.
Is the time difference significant?	No; $p = 0.6387$.
Should White be immediately replaced?	No; validate with a randomized real-world test first.